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Extra Worksheet – Lectures 1 to 4

COMP3670 / COMP6670: Introduction to Machine Learning

Question 1 **Types of learning**

For each of the following tasks, identify which of the three types of learning is involved (supervised, unsupervised, or reinforcement) and describe the data that would be used. If a task can fit more than one type, explain how, and describe the data for each type. Whenever the task is supervised, say what $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{D}$ are.

(a) Recommending a book to a user in an online bookstore.

(b) Playing tic-tac-toe.

(c) Categorising movies into different types.

(d) Learning to play music.

(e) Credit limit: deciding the maximum allowed debt for each bank customer.

Question 2 **Geometry of the perceptron with two features**

Consider the perceptron with $d = 2$ features, so that $\mathbf{x} = (x_1, x_2) \in \mathcal{X} = \mathbb{R}^2$, the weights are $\mathbf{w} = (w_0, w_1, w_2)$, and

$$h_{\mathbf{w}}(\mathbf{x}) = \text{sign}\left(w_1x_1 + w_2x_2 + w_0\right) = \begin{cases} +1, & \text{if } w_1x_1 + w_2x_2 + w_0 > 0, \\ -1, & \text{if } w_1x_1 + w_2x_2 + w_0 \leq 0. \end{cases}$$

(a) Show that the region of the plane where $h_{\mathbf{w}}(\mathbf{x}) = +1$ and the region where $h_{\mathbf{w}}(\mathbf{x}) = -1$ are separated by the line of equation $w_1x_1 + w_2x_2 + w_0 = 0$. Writing this line as $x_2 = ax_1 + b$, find the slope a and the intercept b in terms of w_0, w_1, w_2. Which condition on the weights does this require, and what does the boundary look like when that condition fails?

(b) Using the sign convention above, to which of the two regions do the points *on* the line belong?

(c) Draw a picture of the two regions for $\mathbf{w} = (1, 2, 3)$ and another for $\mathbf{w} = (-1, -2, -3)$. What is the relation between the two hypotheses?

Question 3 **Learning outside the sample**

Let $\mathcal{Y} = \{-1, +1\}$ and let the input space be finite,

$$\mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{x}_{N+1}, \ldots, \mathbf{x}_{N+T}\},$$

so that N inputs appear in the training sample $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$ and the remaining T inputs are never seen. For an unknown target function $f : \mathcal{X} \to \mathcal{Y}$, define the **off-training-set error** of a hypothesis h by

$$E_{\text{off}}(h, f) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{h(\mathbf{x}_{N+t}) \neq f(\mathbf{x}_{N+t})\}.$$

Note that this is the empirical error computed on the T inputs *outside* the sample instead of the N inputs inside it.

(a) Suppose $f(\mathbf{x}) = +1$ for every $\mathbf{x} \in \mathcal{X}$, and

$$h(\mathbf{x}) = \begin{cases} +1, & \text{if } \mathbf{x} = \mathbf{x}_k \text{ with } k \text{ odd and } 1 \leq k \leq N + T, \\ -1, & \text{otherwise.} \end{cases}$$

What is $E_{\text{off}}(h, f)$?

(b) We say that a target function f *generates* $\mathcal{D}$ in a noiseless setting if $y_i = f(\mathbf{x}_i)$ for every $(\mathbf{x}_i, y_i) \in \mathcal{D}$. For a fixed $\mathcal{D}$ with N examples, how many functions $f : \mathcal{X} \to \mathcal{Y}$ generate $\mathcal{D}$ in a noiseless setting?

(c) Fix a hypothesis h and an integer k with $0 \leq k \leq T$. How many of the functions f counted in (b) satisfy $E_{\text{off}}(h, f) = \frac{k}{T}$?

(d) Fix a hypothesis h. If all the functions f that generate $\mathcal{D}$ in a noiseless setting are equally likely, what is the expected off-training-set error $\mathbb{E}_f\big[E_{\text{off}}(h, f)\big]$?

(e) Recall that a learning algorithm $\mathcal{A}$ takes the sample $\mathcal{D}$ as input and returns a hypothesis $\hat{h} = \mathcal{A}(\mathcal{D})$. Argue that for any two learning algorithms $\mathcal{A}_1$ and $\mathcal{A}_2$ that always return the same hypothesis when given the same sample,

$$\mathbb{E}_f\big[E_{\text{off}}(\mathcal{A}_1(\mathcal{D}), f)\big] = \mathbb{E}_f\big[E_{\text{off}}(\mathcal{A}_2(\mathcal{D}), f)\big].$$

You have now proved that, in a noiseless setting, for a fixed $\mathcal{D}$, if all possible target functions are equally likely, then *every* learning algorithm has the same expected error outside the sample. The data alone gives no reason to prefer one prediction over another.

Question 4 **Cross-validation with four folds**

Two candidate models are considered for a binary classification problem with two binary features. $\mathcal{H}_1$ is the set of rules that depend on X_1 only, and $\mathcal{H}_2$ the set of rules that depend on X_2 only. In each case, fitting a model to a set of observations means assigning to each value of the relevant feature the **majority label** of the observations having that value.

The sample below has $N = 16$ observations. Take observations 1 to 4 as the first fold, 5 to 8 as the second, 9 to 12 as the third and 13 to 16 as the fourth. Throughout, use the binary loss.

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
x_1	1	1	0	0	1	1	1	1	0	1	0	0	0	0	1	1
x_2	1	0	1	0	1	0	0	0	0	0	0	0	1	1	0	1
y	1	0	1	0	1	0	0	0	0	0	0	1	1	1	0	0

(a) Fit $\mathcal{H}_1$ and $\mathcal{H}_2$ on all 16 observations, write down the two fitted rules and compute E_{in} for each.

(b) Compute the 4-fold cross-validation error of $\mathcal{H}_1$ and of $\mathcal{H}_2$. Report the four fold errors in each case, as well as their average.

(c) Which model does cross-validation select?

Question 5 **Changing the loss changes what is learned**

Take $\mathcal{Y} = \mathbb{R}$ and let $\mathcal{H}$ be the set of **constant** rules, that is, $h(\mathbf{x}) = c$ for every $\mathbf{x} \in \mathcal{X}$, with $c \in \mathbb{R}$. So a hypothesis is just a single number c: a 'representative' value for the outcomes. Suppose the observed outcomes in the sample are $y_1 \le y_2 \le \cdots \le y_N$.

(a) Suppose the algorithm minimises the in-sample error under the **squared loss**,

$$E_{\text{in}}(c) = \frac{1}{N}\sum_{i=1}^{N}(c - y_i)^2.$$

Writing $\bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i$ for the in-sample mean, show that

$$E_{\text{in}}(c) \;=\; (c - \bar{y})^2 \;+\; \frac{1}{N}\sum_{i=1}^{N}(y_i - \bar{y})^2.$$

The second term does not depend on c. Conclude that the learned hypothesis is $\hat{c}_{\text{mean}} = \bar{y}$, and that it is the *only* minimiser.

[*Hint*: write $c - y_i = (c - \bar{y}) + (\bar{y} - y_i)$ and expand the square.]

(b) Suppose instead the algorithm minimises the in-sample error under the **absolute loss**, defined by $\ell\big((\mathbf{x}, y), h\big) = |y - h(\mathbf{x})|$, so that

$$E_{\text{in}}(c) = \frac{1}{N}\sum_{i=1}^{N}|c - y_i|.$$

(i) Show that for any two numbers $a \le b$ and any c,

$$|c - a| + |c - b| \;\ge\; b - a,$$

with equality if and only if $a \le c \le b$. [*Hint*: check the three cases $c < a$, $a \le c \le b$ and $c > b$ separately.]

(ii) Now pair the observations from the outside in, as (y_1, y_N), (y_2, y_{N-1}), (y_3, y_{N-2}), and so on. Apply (i) to each pair and add the inequalities. Deduce that $E_{\text{in}}(c)$ is smallest exactly when c lies in *every* one of the intervals $[y_i,\ y_{N+1-i}]$.

(iii) Conclude that the learned hypothesis is the in-sample median $\hat{c}_{\text{med}}$, that is, any value such that at least half the observations are at most $\hat{c}_{\text{med}}$ and at least half are at least $\hat{c}_{\text{med}}$. Treat N even and N odd separately, and say in which of the two cases the minimiser is unique.

(c) Suppose the largest observation y_N is replaced by $y_N + \Delta$, where $\Delta \to \infty$, so that this single observation becomes an outlier. What happens to $\hat{c}_{\text{mean}}$ and to $\hat{c}_{\text{med}}$?

Question 6 **Coins, and why verification is not learning**

Recall the bin of marbles of Lecture 3: a coin is tossed N times independently, the probability of heads is μ, and the observed fraction of heads is ν. In the learning analogy, "heads" means "the hypothesis makes a mistake", so $\mu = E_{\text{out}}(h)$ and $\nu = E_{\text{in}}(h)$. Assume we have several coins, each generating its own sample, independently of the others.

For a single coin, the probability of obtaining exactly k heads in N tosses is

$$\mathbb{P}(k \text{ heads in } N \text{ tosses}) = \binom{N}{k} \mu^k (1-\mu)^{N-k}, \qquad \binom{N}{k} = \frac{N!}{k!\,(N-k)!},$$

so that the observed fraction of heads is $\nu = \frac{k}{N}$.

(a) Take $N = 10$ and suppose every coin has $\mu = 0.05$. Compute the probability that *at least one* coin shows $\nu = 0$, when there are 1 coin, 1,000 coins and 1,000,000 coins. Repeat the three computations for $\mu = 0.8$.

(b) Take $N = 6$ and two coins, both with $\mu_1 = \mu_2 = 0.5$. Plot, as a function of $\epsilon \in [0, 1]$, the probability

$$\mathbb{P}\left(\max_{i \in \{1,2\}} |\nu_i - \mu_i| > \epsilon \right),$$

the maximum being taken over the two coins. On the same axes, plot the Hoeffding bound for a *single fixed* coin,

$$\mathbb{P}\big(|\nu - \mu| > \epsilon\big) \le 2e^{-2N\epsilon^2},$$

and the bound obtained by combining Hoeffding with the union bound over $M = 2$ coins, $2Me^{-2N\epsilon^2}$. Comment on how the three curves compare.

(c) Explain, in one or two sentences, what part (a) says about reporting the hypothesis with the best in-sample record, and why the correct bound to use in that situation is the one of part (b) with M equal to the number of hypotheses, and not the single-coin bound.

[*Hint*: It may be easier to compute the probability that *no* coin has the property, and subtract from 1.]

Question 7 **Nested hypothesis sets**

In the polynomial fitting example of Lecture 3, the candidate models formed a nested chain $\mathcal{M}_0 \subset \mathcal{M}_1 \subset \cdots \subset \mathcal{M}_k$, where $\mathcal{M}_q$ is the set of polynomials of degree at most q. Take a single feature, so $\mathcal{X} = \mathbb{R}$ and $\mathcal{Y} = \mathbb{R}$, and consider the two hypothesis sets $\mathcal{H}_2 = \mathcal{M}_2$ and $\mathcal{H}_{10} = \mathcal{M}_{10}$.

(a) Write $\mathcal{H}_2$ and $\mathcal{H}_{10}$ explicitly as sets of functions indexed by their parameters, in the same style as the perceptron hypothesis set of Lecture 1, $\mathcal{H} = \{h_{\mathbf{w}} : w_0, w_1, \ldots, w_n \in \mathbb{R}\}$.

(b) Show that $\mathcal{H}_2 \subset \mathcal{H}_{10}$.

Question 8 **Noise and the out-of-sample error**

Consider a regression problem with the squared loss, in which the outcome is the target function plus noise,

$$Y = f^\star(\mathbf{X}) + \varepsilon, \qquad \mathbb{E}[\varepsilon] = 0, \qquad \mathrm{Var}(\varepsilon) = \sigma^2,$$

with ε independent of $\mathbf{X}$.

(a) Show that, for *any* hypothesis h,

$$E_{\mathrm{out}}(h) \;=\; \sigma^2 \;+\; \mathbb{E}\Big[\big(h(\mathbf{X}) - f^\star(\mathbf{X})\big)^2\Big].$$

(b) Deduce that $E_{\mathrm{out}}(f^\star) = \sigma^2$, and explain why this means that σ^2 is *irreducible*: no hypothesis set, no learning algorithm and no sample size can bring the out-of-sample error below it.

[*Hint for* (a): expand the square inside $E_{\mathrm{out}}(h) = \mathbb{E}\big[(h(\mathbf{X}) - Y)^2\big]$, then use linearity of expectation and the independence of ε and $\mathbf{X}$.]

Question 9 **Error bars for the in-sample and the test error**

A dataset has 600 examples. To properly assess the performance of the final hypothesis, you set aside a randomly selected subset of 200 examples which are never used in the training phase; these form the test set. You use a hypothesis set $\mathcal{H}$ with $M = 1{,}000$ hypotheses and select the learned hypothesis $\hat{h}$ using the remaining 400 training examples. You wish to estimate $E_{\mathrm{out}}(\hat{h})$, and you have two estimates available: $E_{\mathrm{in}}(\hat{h})$, computed on the 400 training examples, and $E_{\mathrm{test}}(\hat{h})$, computed on the 200 test examples that were set aside.

Recall the two bounds from Lecture 3. For a finite $\mathcal{H}$ with M hypotheses, with probability at least $1 - \delta$ and simultaneously for every $h \in \mathcal{H}$,

$$E_{\mathrm{out}}(h) \;\le\; E_{\mathrm{in}}(h) \;+\; \sqrt{\frac{1}{2N}\log\frac{2M}{\delta}},$$

whereas for a hypothesis fixed before the test data is seen, with probability at least $1 - \delta$,

$$\big|E_{\mathrm{test}}(\hat{h}) - E_{\mathrm{out}}(\hat{h})\big| \;\le\; \sqrt{\frac{1}{2N_{\mathrm{test}}}\log\frac{2}{\delta}}.$$

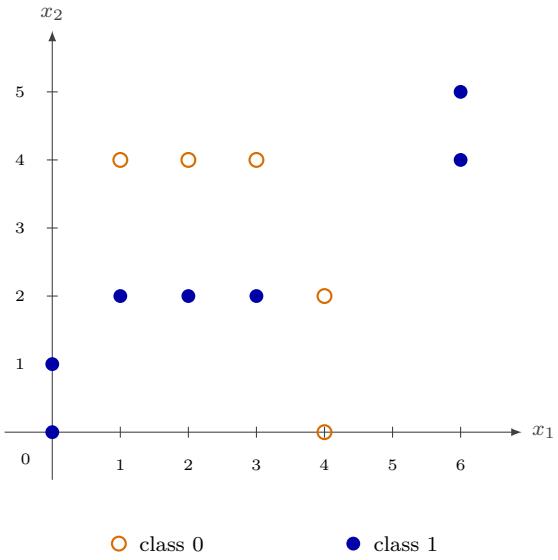
(a) Using a 5% error tolerance ($\delta = 0.05$), compute both error bars for this dataset. Which of the two estimates has the higher error bar?

(b) Notice that the test error bar shrinks as N_{test} grows. Is there then any reason not to reserve even more examples for testing?

(c) Explain why the bound for E_{test} contains no term depending on M, while the bound for E_{in} does.

Question 10 **Growing a classification tree by hand**

The twelve observations below lie in $\mathbb{R}^2$ and carry a binary label.



Grow a CART tree with axis-aligned splits and the **misclassification impurity** $\xi(p, 1-p) = \min(p,\ 1-p)$, continuing until every leaf is pure. Candidate thresholds are the midpoints between consecutive observed values of a feature.

(a) Show that, for this impurity, the impurity drop of a split is exactly the fall in the **number of minority observations**, that is

$$N(R)\,\Delta_R(j,\alpha) \;=\; m(R) \;-\; m\big(R^j_{\alpha,-}\big) \;-\; m\big(R^j_{\alpha,+}\big),$$

where $m(R)$ denotes the number of observations of the less frequent class in the region R.

(b) Grow the tree. At each node, state the split chosen and its impurity drop, and verify that the maximiser is unique. Draw the resulting partition of the plane and the resulting tree.

(c) How many leaves does the tree have, and what is E_{in} of the resulting classifier?

(d) Suppose the stopping rule "no split may leave fewer than $s = 3$ observations in a child" is imposed. Which splits are still allowed, and what is E_{in} of the tree that results?